UNIT 7 · TOPIC 4(A)

Photovoltaic Conversion and Solar-Cell Technology

Syllabus topic: Solar photovoltaic conversion and SPV powered systems

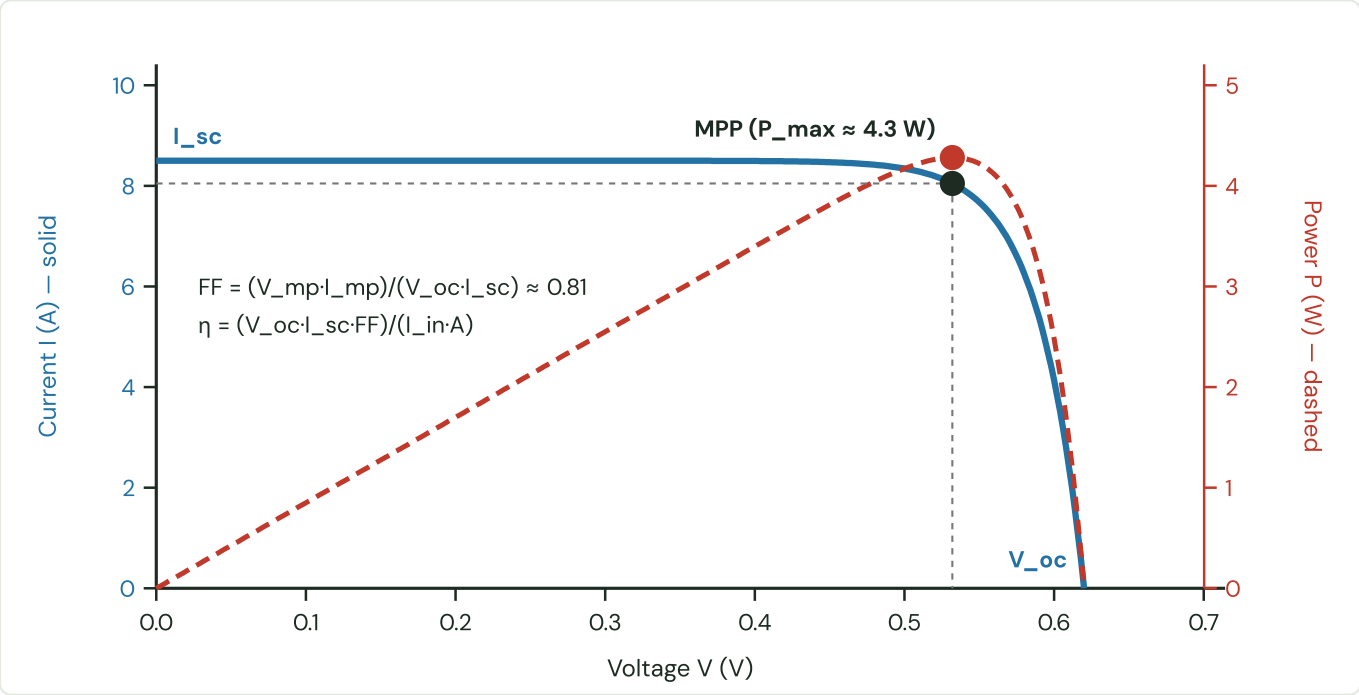
The photovoltaic effect

- PV converts light DIRECTLY into DC electricity through the PHOTOVOLTAIC EFFECT at a semiconductor p–n junction.
- Photons with energy above the BAND GAP (silicon $E_g \approx$ **1.1 eV**) free electron–hole pairs; the junction's built-in field separates them and drives current through the load.
- No moving parts, no intermediate heat engine.

Cell → module → array

- A single silicon CELL gives ~**0.5–0.6 V**.
- Cells in SERIES (adds voltage) form a MODULE (panel); modules in series/parallel form an ARRAY — series adds voltage, parallel adds current.

The I–V curve



Solar-cell I–V (solid) and P–V (dashed) curves, I_{sc} = 8.5 A, V_{oc} = 0.62 V (redrawn from a diode model — the source chart placed the MPP at V_{oc}, where power is zero)

Parameter	Meaning
SHORT-CIRCUIT current I_sc	∝ irradiance and cell area
OPEN-CIRCUIT voltage V_oc	Falls as temperature rises
MAXIMUM POWER POINT (MPP)	Where V·I is greatest — at a voltage below V_oc
FILL FACTOR FF	(V_mp·I_mp)/(V_oc·I_sc)
Efficiency η	(V_oc·I_sc·FF)/(I_in·A)

An **MPPT** charge controller keeps the array at the MPP as sun and temperature change.

Cell types

Type	Features
MONOCRYSTALLINE silicon	Highest efficiency; uniform black
POLY/MULTI-crystalline	Slightly lower efficiency, cheaper; blue speckled
THIN-FILM (a-Si, CdTe, CIGS)	Lower efficiency, but cheap, flexible, better in heat/low light
PERC / TOPCon / HJT; tandem (perovskite–silicon)	Emerging high-efficiency cells (below)

- Output falls with rising cell temperature (~**0.3–0.5%/°C**) and with dust/shading — cleaning matters in Indian fields.
- Rated at **Standard Test Conditions (STC): 1000 W/m², 25 °C, AM1.5**.

Updated information — cell/module technology (as given in the source notes, July 2026)

Item	Value
Commercial module efficiency	PERC ≈ **21–22%**; TOPCon ≈ **22–23%** (market leader, cells ~26%); HJT ≈ 22–27% at cell level
Premium panels	LONGi Hi-MO X6 (TOPCon) 23.0%; Jinko Tiger Neo 22.8%; REC Alpha Pure-R (HJT) 22.3%
Lab records	Best silicon-only cell 27.8% (LONGi HIBC); silicon HJT 27.09%; industrial TOPCon 26.66%
Single-junction limit	Shockley–Queisser ~**33.7%**
PEROVSKITE–SILICON TANDEM	~**34.8%** certified (JinkoSolar/LONGi) — beats the single-junction limit; tandem modules expected commercially from ~2027
Trend	TOPCon has largely replaced PERC in new production; HJT and tandem are the next wave
Thin-film (CdTe, First Solar)	Durable — 88% output retained after 25 yr at NREL; heat-tolerant

> Exam tip: know the ORDER mono > poly > thin-film for efficiency, the STC definition, and that tandem cells beat the single-junction (~33.7%) limit.

Practice MCQs

Q50. PV converts sunlight into:

(a) Heat then electricity
(b) DC electricity directly (photovoltaic effect)
(c) AC directly
(d) Mechanical energy

Q51. The photovoltaic effect occurs at a semiconductor:

(a) Resistor
(b) p–n junction
(c) Capacitor
(d) Transformer

Q52. The band gap of silicon is about:

(a) 0.1 eV
(b) 1.1 eV
(c) 5 eV
(d) 12 eV

Q53. A single silicon cell produces about:

(a) 0.05 V
(b) 0.5–0.6 V
(c) 5 V
(d) 12 V

Q54. Connecting cells in SERIES increases the:

(a) Current
(b) Voltage
(c) Area
(d) Efficiency

Q55. Connecting modules in PARALLEL increases the:

(a) Voltage
(b) Current
(c) Band gap
(d) FF

Q56. Open-circuit voltage V_{oc} of a PV cell as temperature rises:

(a) Increases
(b) Decreases
(c) Constant
(d) Doubles

Q57. Short-circuit current I_{sc} is mainly proportional to:

(a) Temperature
(b) Irradiance and cell area
(c) Voltage
(d) Band gap

Q58. Fill factor FF equals:

(a) $V_{oc} \cdot I_{sc}$
(b) $(V_{mp} \cdot I_{mp})/(V_{oc} \cdot I_{sc})$
(c) I_{sc}/V_{oc}
(d) $V_{mp}+I_{mp}$

Q59. The point of maximum V·I on the I–V curve is the:

(a) V_{oc}
(b) I_{sc}
(c) Maximum power point (MPP)
(d) Origin

Q60. An MPPT controller:

(a) Cools the panel
(b) Keeps the array operating at its maximum power point
(c) Cleans dust
(d) Tracks the sun mechanically

Q61. PV modules are rated at Standard Test Conditions:

(a) 500 W/m², 40 °C
(b) 1000 W/m², 25 °C, AM1.5
(c) 1367 W/m², 0 °C
(d) 800 W/m², 20 °C

Q62. The highest-efficiency common crystalline cell type is:

(a) Monocrystalline silicon
(b) Polycrystalline
(c) Amorphous
(d) CdTe

Q63. Thin-film PV (a-Si, CdTe, CIGS) is characterised by:

(a) Highest efficiency
(b) Lower efficiency but cheap and flexible
(c) No degradation
(d) Needing tracking

Q64. In 2026, commercial TOPCon modules reach efficiencies of about:

(a) 10–12%
(b) 15–17%
(c) 22–23%
(d) 40%

Q65. PERC modules in 2026 deliver about:

(a) 5%
(b) 21–22%
(c) 30%
(d) 45%

Q66. PV output typically falls with rising cell temperature by about:

(a) 0.01%/°C
(b) 0.3–0.5%/°C
(c) 5%/°C
(d) 20%/°C

Answers

Q50 — (b) DC electricity directly (photovoltaic effect)
Q51 — (b) p–n junction
Q52 — (b) 1.1 eV — *Si band gap 1.1 eV.*
Q53 — (b) 0.5–0.6 V — *Cell 0.5–0.6 V.*
Q54 — (b) Voltage
Q55 — (b) Current
Q56 — (b) Decreases — *V_oc falls with temperature.*
Q57 — (b) Irradiance and cell area — *I_sc ∝ irradiance.*
Q58 — (b) (V_mp·I_mp)/(V_oc·I_sc)
Q59 — (c) Maximum power point (MPP)
Q60 — (b) Keeps the array operating at its maximum power point
Q61 — (b) 1000 W/m², 25 °C, AM1.5 — *STC = 1000 W/m², 25 °C, AM1.5.*
Q62 — (a) Monocrystalline silicon
Q63 — (b) Lower efficiency but cheap and flexible
Q64 — (c) 22–23% — *As given in the source notes (July 2026) — TOPCon modules 22–23% (cells ~26%).*
Q65 — (b) 21–22% — *PERC 21–22%.*
Q66 — (b) 0.3–0.5%/°C — *Temperature coefficient ~0.3–0.5%/°C.*

Sources: Sukhatme & Nayak — Solar Energy; Duffie & Beckman — Solar Engineering of Thermal Processes; Rai — Non-Conventional Energy Sources; NREL/ITRPV and manufacturer datasheets (efficiency figures as given in the source notes, July 2026); MNRE guidelines; NPTEL solar-energy lectures.